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The Incidence and Natural Progression of New-Onset Postoperative Atrial Fibrillation

Sahar S. Abdelmoneim, MD, MS, Emelie Rosenberg, MD, Marcella Meykler, BA, Bimal Patel, MD, Bharath Reddy, MD, Jean Ho, BA, Igor Klem, MD, Jaspal Singh, BA, Berhane Worku, MD, Robert F. Tranbaugh, MD, Terrence J. Sacchi, MD, John F. Heitner, MD

ABSTRACT

OBJECTIVES This study aimed to characterize the natural progression and recurrence of new-onset postoperative atrial fibrillation (POAF) during an intermediate-term follow-up post cardiac surgery by using continuous event monitoring.

BACKGROUND New-onset POAF is a common complication after cardiac surgery and is associated with an increased risk for stroke and all-cause mortality. Long-term data on new POAF recurrence and anticoagulation remain sparse.

METHODS This is a single-center, prospective observational study evaluating 42 patients undergoing cardiac surgery and diagnosed during indexed admission with new-onset, transient, POAF between May 2015 and December 2019. Before discharge, all patients received implantable loop recorders for continuous monitoring. Study outcomes were the presence and timing of atrial fibrillation (AF) recurrence (first, second, and more than 2 AF recurrences), all-cause mortality, and cerebrovascular accidents. A "per-month interval" analysis of proportion of patients with any AF recurrence was assessed and reported per period of follow-up time. Kaplan-Meier analysis was used to calculate the time to first AF recurrence and report the first AF recurrence rates.

RESULTS Forty-two patients (mean age 67.6 ± 9.6 years, 74% male, mean $CHADS_2$-VASc 3.5 ± 1.5) were evaluated during a mean follow-up of 1.7 ± 1.2 years. AF recurrence after discharge occurred in 30 patients (71%) and of those, 59% had AF episodes equal to or longer than 5 minutes (median AF duration at 1 month was 32 minutes [interquartile range 5.5–106], whereas median AF duration beyond 1 month was 15 minutes [interquartile range 6.3–49]). Twenty-four (80%) of the 30 patients had their first AF recurrence within the first month. During months 1 to 12 follow-up, 76% of patients had any AF recurrences (10% had their first AF recurrence, 43% had their second AF recurrence, and 23% had more than 2 AF recurrences). Beyond 1 year of follow-up, 30% of patients had any AF recurrences (10% had their first AF recurrence, 7% had their second AF recurrence, and 13% had more than 2 AF recurrences). Using Kaplan-Meier analysis, the median time to first AF recurrence was 0.83 months (95% CI: 0.37 to 6) and the detection of first AF recurrence rate at 1, 3, 6, 12, 18, and 24 months was 57.1%, 59.5%, 64.3%, 64.3%, 67.3%, and 73.2%, respectively. During follow-up, there was 1 death ([−] AF recurrence) and 2 cerebrovascular accidents ([+] AF recurrence).

CONCLUSIONS In this study of continuous monitoring with implantable loop recorders, the recurrence of AF in patients who develop transient POAF is common in the first month postoperatively. Of the patients who developed postoperative AF, 76% had any recurrence in months 1 to 12, and 30% had any recurrence beyond 1-year follow-up. Current guidelines recommend anticoagulation for POAF for 30 days. The results of this study warrant further investigation into continued monitoring and longer-term anticoagulation in this population within the context of our findings that AF duration was <30 minutes beyond 1 month. (J Am Coll Cardiol EP 2021;7:1134–1144) © 2021 by the American College of Cardiology Foundation.

From the Division of Cardiology, Weill Cornell Medicine, New York-Presbyterian Brooklyn Hospital, Brooklyn, New York, USA.
The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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The occurrence of new-onset postoperative atrial fibrillation (POAF) after cardiac surgery is reported to be 20% to 62% in patients without prior atrial fibrillation (AF) (1–4). Although new-onset POAF was once believed to be self-limiting due to spontaneous restoration of sinus rhythm before hospital discharge, several studies have shown a strong association with cardiovascular events, including a 2-fold increase in 30-day mortality, a 3-fold higher 30-day stroke risk (2,5,6), prolonged hospitalization, increase in hospital readmissions (7), and greater health care costs (3,8).

Initiation of oral anticoagulation (OAC) therapy for new-onset POAF varies between the different surgical and clinical practices. The decision on when to discontinue OAC remains controversial in such a population. Current guidelines for the management of POAF after cardiac surgery recommend anticoagulation for at least 30 days after conversion to sinus rhythm (9–11). Indeed, studies have shown the treatment of POAF can lower the risk of stroke by 65% to 80% (11). Despite these clinical practice guidelines, it was reported that only 14% of patients with new-onset POAF were discharged on OAC (12).

Long-term data on POAF recurrence remains sparse and controversies remain regarding the true incidence after discharge from the hospital (6). A cardiac rehabilitation registry including all patients discharged after cardiac surgery reported POAF recurrence rate of 10.7% with reported incidence varying based on type of surgery (6.6% coronary, 21.9% valve, and 14.3% combined surgery) (13). Another study using 7-day Holter monitoring, along with intermittent electrocardiograms (ECGs), in postoperative patients undergoing open heart surgery showed late (beyond 3 months) POAF recurrence rate of 44% at a mean follow-up duration of 3.6 years (14).

There has been variability in reporting of POAF on its natural progression regarding recurrence (13). Such variability in reported recurrence rate reflects the difference in population characteristics, the asymptomatic nature of the arrhythmia, type of surgery, methodology used for rhythm monitoring (either single 12-lead ECG at follow-up, continuous Holter monitoring, or short-term event recorder), variability of follow-up duration, and variability in time points for defining POAF recurrence.

Given the need to effectively establish a surveillance and treatment strategy for POAF, our study aimed to characterize the natural progression of new-onset POAF, including rate of AF recurrence during an intermediate-term follow-up postsurgery using a continuous event monitor (implantable loop recorder [ILR]) to determine the need for long-term anticoagulation beyond the current recommendation of 30 days.

METHODS

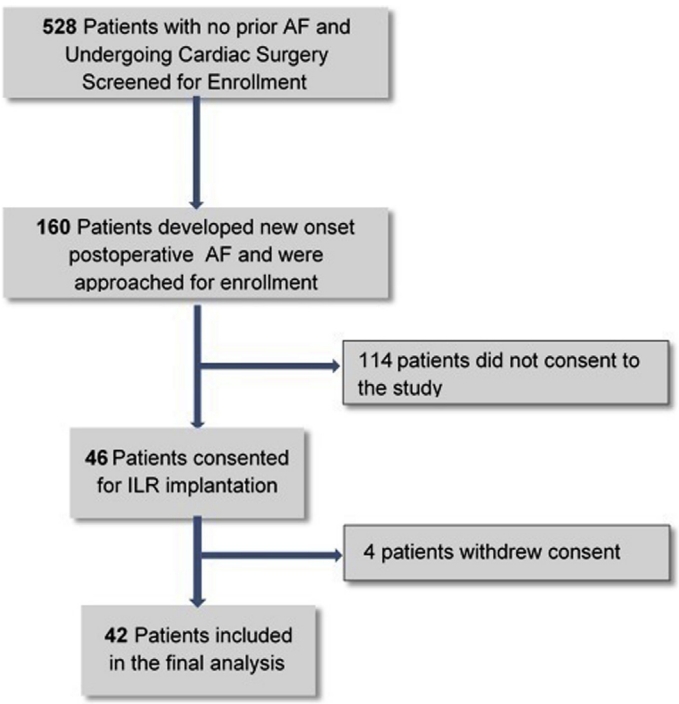
This is a single-center prospective cohort study. The study was approved by the Institutional Review Board of New York-Presbyterian Brooklyn Methodist Hospital, and all patients provided a written informed consent to the study.

STUDY POPULATION. Patients older than 18 years and undergoing open heart surgery (coronary artery bypass surgery [CABG] or valve surgery) between May 2015 and December 2019 were identified from the cardiothoracic operative schedule and screened for study eligibility. Inclusion criteria included those who developed new-onset POAF during the initial postoperative period (defined as newly diagnosed AF occurring up to 10 days after surgery as documented by ECG or continuous telemetry monitoring for a minimum of 30 seconds, with no prior history of AF) and successfully reverted (either by

ABBREVIATIONS AND ACRONYMS

AF = atrial fibrillation
CABG = coronary artery bypass surgery
CVA = cerebrovascular accident
ECG = electrocardiogram
ILR = implantable loop recorders
IQR = interquartile range
OAC = oral anticoagulation
POAF = postoperative atrial fibrillation

FIGURE 1 Study Cohort



Overall study flow diagram. AF= atrial fibrillation. ILR= implantable loop recorder.

TABLE 1 Clinical Characteristics of All Study Population as well as Categorized by Atrial Fibrillation (AF) Recurrence Postoperatively (N = 42)

	Total Population	AF Recurrence (n = 30)	No AF Recurrence (n = 12)	p Value
Age (y)	67.6 ± 9.6	66.7 ± 9.9	68.6 ± 8.3	0.56
Male	31 (74)	21 (70)	9 (75)	0.45
Race				0.46
Caucasian	14 (33)	9 (30)	5 (42)	
African American	13 (31)	11 (37)	2 (17)	
Hispanic	1 (2)	1 (3)	0 (0)	
Other	13 (31)	9 (30)	4 (33)	
BMI (kg/m^2)	29.9 ± 7.8	31.5 ± 8.7	26.5 ± 5.3	0.03
Diabetes	16 (38)	9 (30)	7 (58)	0.10
Hypercholesterolemia	33 (79)	25 (84)	8 (67)	0.45
Hypertension	37 (88)	29 (96.7)	8 (67)	0.03
Current smoking	7 (17)	6 (21)	1 (8)	0.34
History of cancer	4 (9.5)	4 (13)	0 (0)	0.18
CAD	23 (55)	15 (50)	8 (67)	0.38
MI	7 (17)	3 (10)	4 (33)	0.08
COPD	4 (10)	3 (10)	1 (8)	0.84
OSA	2 (5)	2 (7)	0 (0)	0.35
PAD	22 (52)	12 (40)	10 (83)	0.01
CHF	17 (41)	12 (40)	5 (42)	0.9
$CHADS_2$ VASCs	3.5 ± 1.5	3.3 ± 1.52	3.8 ± 1.53	0.36
% Stroke risk	4.1 ± 2.3	3.9 ± 2.2	4.8 ± 2.5	0.31
Operative details				
Valve surgery alone	14 (33)	11 (37)	3 (25)	0.35
CABG alone	26 (62)	17 (57)	8 (67)	0.21
Combined CABG and valve surgery	2 (5)	2 (7)	0 (0)	0.09
Aortic clamp time (min)	89.5 ± 26.1	96.3 ± 27.8	75.6 ± 15.5	<0.01
Pump time (min)	119.6 ± 32.1	128. ± 32.7	99.3 ± 20	<0.01
Mean postoperative hospital stay (d)	9.7 ± 4.7	9.8 ± 5.1	9.5 ± 4.1	0.83
Medications				
Outpatient beta-blocker use	28 (67)	22 (73)	8 (67)	0.52
Patients discharged on beta blockers	39 (93)	29 (97)	10 (83)	0.41
Amiodarone prophylaxis (post-surgery but prior AF development)	10 (24)	8 (27)	2 (17)	0.4
Patients discharged on amiodarone	32 (76)	23 (77)	9 (75)	0.76
Patients discharged on oral anticoagulant	24 (57)	21 (70)	3 (25)	0.005 <0.01
INR at discharge	1.6 ± 0.57	1.75 ± 0.6	1.25 ± 0.25	<0.01
In-hospital labs				
WBC (× 10^9/l)	11.5 ± 5.6	10.2 ± 4.9	15.3 ± 6.1	0.02
Hemoglobin (g/dl)	11.1 ± 2.4	11 ± 2.3	11.1 ± 2.5	0.91
Platelets	169.7 ± 54.9	163.3 ± 51.1	180 ± 64.2	0.44
BUN (mg/dl)	20.1 ± 8.6	20.7 ± 9.7	19.3 ± 5.0	0.57
Creatinine (mg/dl)	1.3 ± 1.1	1.3 ± 1.0	1.5 ± 1.3	0.42
Na (mEq/l)	141.5 ± 3.5	142.7 ± 3.1	139.1 ± 3.2	<0.01
K (mEq/l)	4.2 ± 0.49	4.2 ± 0.49	4.5 ± 0.42	0.06
TSH (μg/dl)	1.8 ± 1.6	1.7 ± 1.0	1.7 ± 0.9	0.98
Preop pBNP(pg/ml)	312 (145-1,783)	274 (151.5-1,472)	1,628 (112-15,689)	0.26
Postop pBNP (pg/ml)	895 (127-2,891)	423 (125-2,131)	3,117 (1,409-20,560)	0.14

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spontaneous, pharmacologic, or electrical cardioversion) to normal sinus rhythm before discharge from the hospital. Patients with persistent new-onset POAF that lasted at least 7 days postoperative and had been cardioverted to sinus rhythm (spontaneously, medically, or electrically) were included. Study exclusion criteria included prior history of AF.

DATA COLLECTION AND MONITORING. Eligible and consenting patients who were discharged from the hospital and in normal sinus rhythm underwent an ILR implantation before discharge (15) (Reveal LINQ Implantable Cardiac Monitoring System; Medtronic, Inc., Mountain View, California) or a pacemaker/implantable cardiac defibrillator if indicated. Implantation was performed per manufacturer's guidelines using a minimal invasive subcutaneous implantation technique. Implantation was performed in the electrophysiology laboratory (16,17).

Data collection included medical history (clinical demographics, cardiovascular risk factors, medications including amiodarone and beta blockers), indications for open heart surgery, and details of operation. The decision to use amiodarone by the clinical team was based on high-risk patients (i.e., pre-existing heart failure, age >75 years, obesity, and chronic renal failure). In general, the surgical team's protocol for stopping amiodarone postoperatively is 6 weeks; however, such decision was individualized per patient. Hence, such data were collected both at time of enrollment and during follow-up via phone calls, interviews, and from electronic health records. Data from echocardiographic testing was also collected.

FOLLOW-UP AND STUDY OUTCOMES ASCERTAINMENT. Patients were prospectively followed up via electrophysiology device checks both in clinic and via remote monitoring. Device interrogation was performed at 0 to 1 month, 1 to 2 months, 2 to 3 months, 3 to 6 months, 6 to 12 months, and beyond 12 months postoperatively. ILR data were transmitted to a care link database and reports of device interrogation were stored in the medical records at each time point. ILR data were interrogated by the clinical electrophysiologist on service who was blinded to our study. ILR was explanted at the discretion of the treating physician per standard of care.

The study outcomes included the identification of POAF recurrence as detected by the ILR and time to recurrence. Recurrence of POAF was defined as any

documented AF episode (>30 seconds' duration) on the monitoring devices following hospital discharge and during study follow-up period. Recurrence of POAF post hospital discharge was further categorized to first AF recurrence, second AF recurrence, and more than 2 AF recurrences. First AF recurrence was defined as the proportion of patients experiencing first-time AF recurrence during follow-up. Second AF recurrence was defined as the proportion of patients experiencing a second AF recurrence after the first AF recurrence during follow-up. More than 2 AF recurrences was defined as the proportion of patients experiencing more than 2 AF recurrences during follow-up. AF burden (expressed as a percentage) and duration of AF episodes (in minutes) were also reported. In addition, we reported on the long-term events of all-cause mortality and/or cerebrovascular accidents (CVAs) during follow-up.

STATISTICAL ANALYSIS. For continuous variables, mean ± standard deviation were used if the data were normally distributed, whereas median and ranges were applied for skewed data. For categorical variables, numbers and percentage were used. Baseline characteristics were presented in tabular form for the population, and in subgroups defined by anticoagulation status at baseline (before discharge from the hospital). The intergroup comparisons were performed by independent Student's *t*-test continuous variables and a chi-square test for categorical variables, as considered appropriate. The study outcomes included the identification of POAF recurrence as detected by the ILR. A "per-month interval" analysis of proportion of patients with any AF recurrence was assessed and reported per period of follow-up time (0-1 month, 1-2 months, 2-3 months, 3-6 months, 6-12 months, and beyond 12 months postoperatively). In addition, Kaplan-Meier survival analysis was used to calculate the time to first AF recurrence and report the first AF recurrence rates. All statistical analyses were performed with JMP software version 10 (SAS Institute Inc.). All probability values were 2-sided. Statistical significance was set a priori at 2-tailed $p < 0.05$.

RESULTS

STUDY POPULATION AND PROCEDURAL CHARACTERISTICS. In this single-center prospective study, a total of 528 patients undergoing cardiac surgery with no prior history of AF were prospectively identified by screening the open heart surgery operative schedule (CABG, valve, or combined surgeries) during the time frame between May 2015 and December 2019. Of these, 204 patients (38.6%) were treated with prophylactic amiodarone. Among the AF-naive cardiac surgery patients, 160 (30%) patients developed new-onset POAF. During the study period, enrollment was interrupted due to research personnel loss. Hence, 46 (29%) of 160 patients consented for our study and underwent ILR implantation for long-term arrhythmia monitoring. Four patients withdrew their consent before ILR implantation, thus a total of 42 patients were included in the final analysis. Figure 1 illustrates the study population flow chart.

TABLE 1 Continued

	Total Population	AF Recurrence (n = 30)	No AF Recurrence (n = 12)	p Value
Preoperative echo data				
LVEF (%)	52.03 ± 13.6	53.4 ± 13.1	48.5 ± 15	0.30
HFrEF	11 (26)	6 (0)	5 (42)	0.13
Dilated LA (mild or moderate or severe)	13 (30)	7 (23)	6 (50)	0.08
Degree of dilated LA				0.12
Mild	6 (14)	3 (10)	3 (25)	
Moderate	2 (5)	2 (7)	0 (0)	
Severe	5 (13)	2 (7)	3 (25)	
Postoperative echo data				
LVEF (%)	40.6 ± 15.9	43.1 ± 15.2	27.7 ± 11.6	0.13
Dilated LA	8 (19)	6 (20)	2 (17)	0.39
Degree of dilated LA				0.62
Mild	5 (12)	4 (13)	1 (8)	
Moderate	0 (0)	0 (0)	0 (0)	
Severe	3 (7)	2 (7)	1 (8)	

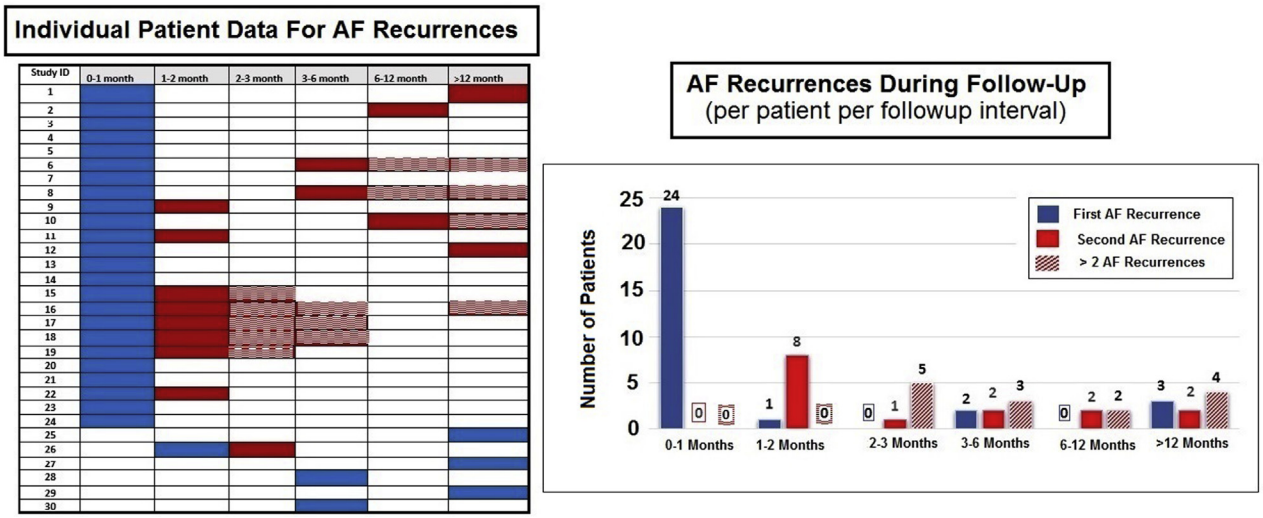
Values are mean ± SD, or n (%), or median (25% and 75% Interquartile range).

AF = atrial fibrillation; BMI = body mass index; BUN = blood urea nitrogen; CABG = coronary artery bypass graft; CAD: coronary artery disease; CHF = congestive heart failure; COPD = chronic obstructive pulmonary disease; HFrEF = heart failure reduced ejection fraction (defined as LVEF ≤45%); INR = international normalized ratio; K = Potassium; LA = left atrium; LVEF = left ventricular ejection fraction; MI = myocardial infarction; Na = Sodium; OSA = obstructive sleep apnea; PAD = peripheral arterial disease; pBNP = pro brain natriuretic peptide; TSH = thyroid-stimulating hormone; WBC = white blood cells.

Table 1 illustrates the baseline demographic and clinical characteristics of the study population. Mean age of all patients was 67.6 ± 9.6 years. Patients were predominantly male (74%) and the population was equally distributed between African American (31%) and Caucasian (33%) individuals. CABG was performed in 26 (62%) patients, valve replacement or repair in 14 (33%), and both in 2 (5%). Most valve replacements (93%) were bioprosthetic valves. Mean postoperative hospital stay was 9.7 ± 4.7 days. Intraoperative complications occurred in 1 patient who developed a left-sided hemothorax post CABG.

All patients received a long-term monitoring device implantation within 9.2 ± 7.6 days of surgery (median 7 days; 25%-75% interquartile range [IQR]: 6-10 days). Monitoring device included Loop recorder

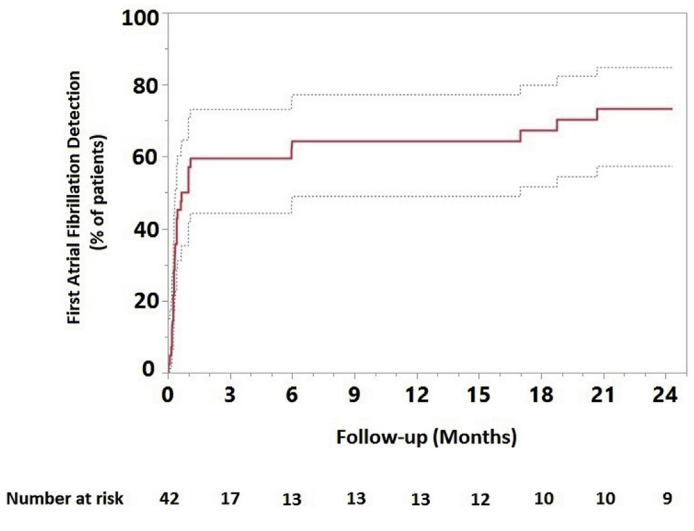
FIGURE 2 AF Recurrences in Study Cohort



(Left) Individual patient data across follow-up time points for all atrial fibrillation (AF) recurrences (N = 30). First AF recurrence **(blue)** represents patients experiencing first-time AF recurrence at any follow-up time point. Second AF recurrence **(red color)** represents patients experiencing a second AF recurrence after the first AF recurrence at any follow-up time point. More than 2 AF recurrences **(striped red color)** represents patients experiencing more than 2 AF recurrences at any follow-up time point. **(Right)** Bar graph extrapolated form the individual patient data, illustrating AF recurrence during follow-up (per patient per follow-up time points). Color coding and labels are shown in the graph.

(Reveal LINQ Insertable Cardiac Monitoring System, Medtronic, Inc.) implantation in 40 patients (95%), pacemaker implantation in 1 patient, and implantable cardiac defibrillator in 1 patient.

FIGURE 3 Incidence of AF Recurrence From Kaplan-Meier Analysis

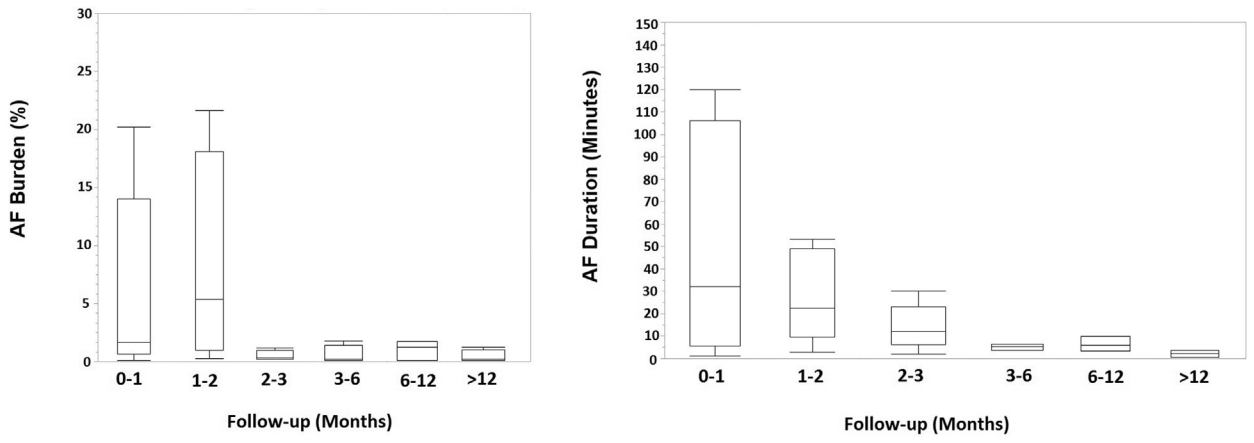


Time to first detection of AF recurrence. **Dotted lines** represent 95% CIs. AF= atrial fibrillation.

POAF (IN-HOSPITAL NATURAL HISTORY). New-onset POAF occurred within a mean of 2.85 ± 1.9 days from surgery, and palpitations were reported in 7 (17%) of these patients. Detection of new-onset POAF, during in-hospital stay, was documented on ECG in 6, telemetry in 4, or both in 31. Resolution of new-onset POAF episodes in-hospital, occurred spontaneously in 3 (7%) patients, with amiodarone /antiarrhythmic medications in 37 (88%) patients, and with amiodarone and electrical cardioversion in 2 (5%) patients. All patients were in normal sinus rhythm at time of discharge.

STUDY OUTCOMES. AF recurrence and symptomatology. AF recurrence after discharge occurred in 30 patients (71%). Baseline characteristics of patients with and without AF recurrence are shown in Table 1. Patients with AF recurrence were more likely to have a high body mass index, increased hypertension, aortic clamp time, and longer pump time. Figure 2 illustrates the breakdown of all AF events in the study cohort across different follow-up time points. First AF recurrence after discharge occurred in 24 (80%) patients within the first month, and in 3 (10%) patients during months 1 to 12. Second AF recurrence occurred in 13 (43%) patients, whereas more than 2 AF recurrences occurred in 7 (23%) patients during months 1 to 12.

FIGURE 4 AF Burden (%) and Duration (Minutes)



Box-and-whisker plot showing AF burden (%), and duration (minutes) during follow-up intervals. The **center line** indicates the median; the **bottom and the top of the box**, the 25th and 75th percentiles, respectively. AF= atrial fibrillation.

Beyond 1 year of follow-up, 30% of patients had any AF recurrences (10% had their first AF recurrence, 7% had their second AF recurrence, and 13% had more than 2 AF recurrences) (Figure 2). Using Kaplan-Meier survival analysis, the median time to first AF recurrence was 0.83 months (95% CI: 0.37 to 6) and the detection of first AF recurrence rate at 1, 3, 6, 12, 18, and 24 months was 57.1%, 59.5%, 64.3%, 64.3%, 67.3%, and 73.2%, respectively (Figure 3).

Any symptom of palpitation during the entire follow-up was reported in 12 (29%) of the 42 patients. However, in those with AF recurrence, palpitation was reported in 6 (25%) of 24 patients with AF at 0 to 1 month, 3 (33%) of 9 patients with AF at 1 to 2 months, 2 (33%) of 6 patients with AF at 2 to 3 months, 2 (29%) of 7 patients with AF at 3 to 6 months, 1 (25%) of 4 patients with AF at 6 to 12 months, and 1 (11%) of 9 patients with AF at >12 months.

AF burden and duration. AF burden and episode duration across different follow-up time points are displayed in Figure 4. In patients with AF recurrence who experienced palpitation, AF burden was not significantly higher compared with those with AF recurrence with no palpitation (median 7%, interquartile range [IQR] 0.85-16.6 vs. median 0.9%, IQR 0.25-10.5; p = 0.293).

ILR-recorded AF episodes with duration equal to or longer than 5 minutes occurring during the entire follow-up period were reported in 16 (59%) of 30 patients (median AF duration at 1 month was 32 min [IQR: 5.5-106], whereas median AF duration beyond 1 month was 15 minutes [IQR 6.3-49]). Of those 16 patients, the symptom of palpitation was reported in 4 patients (25%) only. Patients with AF duration lasting ≥5 minutes were older, men, hypertensive, hypercholesteremic, with prior myocardial infarction, obstructive sleep apnea, most underwent valve surgery, and reported higher use of amiodarone, beta blockers, and OAC compared with those with AF duration <5 min, although the differences did not reach statistical significance (data not shown).

Antiarrhythmics and oral anticoagulants. Antiarrhythmics and OAC treatment were initiated and or stopped based on individualized clinical decision at the discretion of cardiothoracic surgeon and cardiologist. Of the overall 42 patients included in the final analysis, amiodarone was prescribed to 32 (76%) patients at 0 to 1 month, 19 (45%) at 1 to 2 months, 19 (45%) at 2 to 3 months, 8 (19%) at 3 to 6 months, 3 (7%) at 6 to 12 months, and 3 (7%) patients at >12 month. However, in those with AF recurrence, amiodarone was given to 20 (83%) of 24 patients at 0 to 1 month, 5 (56%) of 9 at 1 to 2 months, 3 (50%) of 6 at 2 to 3 months, 2 (29%) of 7 at 3 to 6 months, 1 (25%) of 4 at 6 to 12 months, and 3 (33%) of 9 at >12 months.

Beta blockers were given to 39 (93%) of 42 patients within first 3 months and maintained thereafter during follow-up with no substantial changes (modest adjustment in doses: in 18 patients at 3-6 months, in 5 patients at 6-12 months, and in 1 patient at >12 months).

OACs were prescribed to 24 (57%) of 42 patients within the first 3 months (all being on Coumadin), 13 (31%) patients at 3 to 6 months, 11 (26%) at 6 to

TABLE 2 Clinical Characteristics of Those Experiencing First AF Recurrence at 30 Days and Beyond 30 Days

	First AF Recurrence at 1 Month (n = 24)	First AF Recurrence Beyond 1 Month (n = 6)	p Value
Age (yr)	66.6 ± 10.6	68.3 ± 7.0	0.64
Male	17 (71)	4 (67)	0.84
Race			0.21
Caucasian	9 (38)	0 (0)	
African American	8 (33)	4 (67)	
Hispanic	1 (4)	0 (0)	
Others	6 (25)	2 (33)	
BMI (kg/m^2)	31.8 ± 8.5	29.2 ± 9.4	0.56
Diabetes	5 (21)	5 (83)	<0.01
Hypercholesterolemia	20 (83)	5 (83)	0.45
Hypertension	23 (96)	6 (100)	0.61
Current smoking	5 (21)	1 (17)	0.81
History of cancer	4 (17)	0 (0)	0.28
CAD	12 (50)	4 (67)	0.46
MI	2 (8)	1 (17)	0.54
COPD	2 (8)	1 (17)	0.54
OSA	2 (8)	0 (0)	0.46
PAD	9 (38)	4 (67)	0.19
CHF	10 (42)	4 (67)	0.27
CHADS2 VASCs	3.2 ± 1.5	4.5 ± 1.6	0.11
% Stroke risk	3.7 ± 2.2	5.6 ± 3.2	0.21
Operative details			
Valve surgery alone	12 (50)	1 (17)	0.12
CABG alone	12 (50)	8 (83)	0.14
Combined CABG and valve surgery	2 (8)	0 (0)	0.86
Aortic clamp time (min)	96.3 ± 29.1	83.8 ± 17.8	0.14
Pump time (min)	131.8 ± 34.2	108 ± 15.9	0.02
Mean postoperative hospital stay (days)	8.8 ± 3.6	14.7 ± 7.4	0.11
Medications			
Outpatient beta-blocker use	18 (75)	4 (67)	0.79
Patients discharged on beta blockers	23 (96)	6 (100)	0.47
Amiodarone prophylaxis (postsurgery but prior AF development)	5 (22)	3 (50)	0.17
Patients discharged on amiodarone	19 (79)	5 (83)	0.82
Patients discharged on oral anticoagulants	15 (63)	1 (17)	0.04
INR at discharge	1.9 ± 0.6	1.1 ± 0.04	<0.01
In-hospital labs			
WBC (· 10^9/l)	10.6 ± 5.2	8.9 ± 2.9	0.29
Hemoglobin (g/dl)	11.2 ± 2.7	10.5 ± 1.7	0.39
Platelets	160.5 ± 46.4	165.7 ± 72.6	0.87
BUN (mg/dl)	18.7 ± 5.9	28.8 ± 16.4	0.19
Creatinine (mg/dl)	.97 ± 0.3	2.3 ± 1.9	0.17
Na (mEq/l)	1,42.3 ± 3.1	1,431 ± 4.3	0.74
K (mEq/l)	4.1 ± 0.49	4.5 ± 0.29	0.03
TSH (μg/dl)	1.4 ± 0.7	2.9 ± 1.4	0.35
Preop pBNP (pg/ml), median (25% and 75% IQR)	269 (162-1,107)	82.2 (99.3-45,098)	0.37
Postop pBNP (pg/ml), median (25% and 75% IQR)	403.5 (110-1,156.3)	2236 (844.5-21,328)	0.03

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12 months, and 11 (26%) at >12 months. In those with AF recurrence, OAC was given to 18 (75%) of 24 patients at 0 to 1 month, 6 (67%) of 9 patients at 1 to 2 months, 5 (83%) of 6 patients at 2 to 3 months, 4 (57%) of 7 patients at 3 to 6 months, 2 (50%) of 4 patients at 6 to 12 months, and 3 (33%) of 9 patients at >12 months.

All-cause mortality and CVAs. The mean follow-up was 1.7 ± 1.2 years. All-cause mortality was reported in 1 patient within 6 months of follow-up postsurgery. This patient did not have AF recurrence but had a history of aortic valve replacement, with multiple comorbidities.

CVAs occurred in 2 patients. The first patient had aortic valve replacement and ascending aortic aneurysm repair (discharge International Normalized Ratio of 2.6) and experienced repeated AF recurrence on ILR device interrogation (AF duration <5 min, AF burden of 0.1%) and CVA 1 week postoperatively. The second patient had aortic valve replacement (discharged on OAC), and experienced repeated AF recurrence (within 3-month follow-up, AF duration ≥5 min, AF burden of 29.8%). At 6-month ILR device interrogation, the patient had been AF free for 1 month and hence OAC was discontinued. During 7 months follow-up, the patient was hospitalized with a transient ischemic attack despite ILR interrogation before admission, indicating the patient was free of AF episodes. The patient was discharged on OAC and follow-up ILR interrogation showed no repeated AF recurrence.

SUBGROUP ANALYSIS. Type of surgery. Study population was also categorized based on the type of index surgery (CABG vs. valve surgery) and there was no statistically significant difference between the POAF recurrence between the 2 groups (17 [68%] in CABG vs. 13 (81%) in valve surgery; p = 0.350). However, as expected, patients undergoing CABG were more likely to have prior coronary artery disease, peripheral arterial disease, and increased pump time (minutes), versus those with valve surgery.

First AF recurrence. Patients with first AF recurrence beyond 1 month were more likely to be diabetic, have heart failure with reduced ejection fraction, prolonged pump time, and lower preoperative and postoperative left ventricular ejection fraction compared with those with first POAF recurrence within 1 month (Table 2).

DISCUSSION

The following are the major findings of our current study: 1) recurrence of POAF after discharge and through a mean follow-up period of 1.7 ± 1.2 years occurred in 71% of patients, and of those 59% had AF episodes equal to or longer than 5 minutes (median AF duration at 1 month was 32 min [IQR: 5.5-106 min], whereas median AF duration beyond 1 month was

15 min [IQR: 6.3-49.0 min]); 2) first AF recurrence within the first month of follow-up post discharge occurred in 80% of patients; 3) during months 1 to 12 of follow-up, 76% of patients had any AF recurrences (10% had first AF recurrence, 43% had second AF recurrence, and 23% had more than 2 AF recurrences); 4) beyond 1 year of follow-up, 30 % of patients had any AF recurrences (**Central Illustration**).

The incidence of in-hospital POAF for all patients undergoing open heart surgery in our center was 30%, which is consistent with prior report rates of 20% to 62% (2,6,18). All patients had new-onset POAF in our study within a mean of 2.85 ± 1.9 days post-surgery, which is in agreement with prior published studies that have shown peak onset at postoperative day 2 (19). On average, the first AF recurrence occurred after 26 days (using Kaplan-Meier survival analysis, the median time to first AF recurrence was 0.83 months; 95% CI: 0.37 to 6) and the detection of first AF recurrence rate at 1 month was 57.1%. Two previous studies have also shown peak AF recurrence at 12 ± 6 days (6,18). We have reported that AF recurrence after discharge and through a mean follow-up period of 1.7 years occurred in 71% of patients (and most were not associated with symptoms of palpitation). This observed percentage is slightly higher than that reported by a similar, but smaller study by El-Chami et al (1), who found an occurrence of 60%. This difference may be attributed to the difference of the enrolled population in our study, including both valve surgery and CABG.

In our study, 57% and 76% patients were discharged on OAC and amiodarone, respectively. This management decision was individualized per patient and the decision was solely at the discretion of the multidisciplinary care team (cardiothoracic surgeon and cardiologists) taking into account different individual clinical risk, $CHADS_2$-VASC score, number of AF episodes in hospital postoperatively, risk of bleeding, and noncompliance risk with medications. We have shown that during the first month post-discharge, AF recurrence is the highest (71% of patients, and of those, 59% had AF episodes equal to or longer than 5 min; median AF duration at or beyond 1 month was 32 min [IQR: 5.5-106.0 min] and 15 min [IQR: 6.3-49.0 min], respectively). However, we found that for 1 to 12 months postoperative follow-up, the recurrence is still high (first AF recurrence in 10%, second AF recurrence in 43%, and more than 2 AF recurrences in 23% of patients). These findings are important with respect to the current guidelines by CHEST (American College of Chest Physicians) and Society of Thoracic Surgeons, which recommend OAC in this population for a duration of or beyond 30 days.

TABLE 2 Continued

	First AF Recurrence at 1 Month (n = 24)	First AF Recurrence Beyond 1 Month (n = 6)	p Value
Preoperative echo data			
LVEF (%)	56.2 ± 11.4	42.3 ± 12.9	0.04
HFrEF	3 (13)	3 (50)	0.05
Dilated LA (mild or moderate or severe)	7 (30)	1 (17)	0.51
Degree of dilated LA			0.63
Mild	3 (13)	0 (25)	
Moderate	2 (9)	0 (0)	
Severe	2 (9)	1 (17)	
Postoperative echo data			
LVEF (%)	49.9 ± 11.1	29.6 ± 13.5	0.02
Dilated LA	5 (50)	1 (20)	0.26
Degree of dilated LA			0.44
Mild	3 (30)	1 (20)	
Moderate	0 (0)	0 (0)	
Severe	2 (20)	0 (0)	

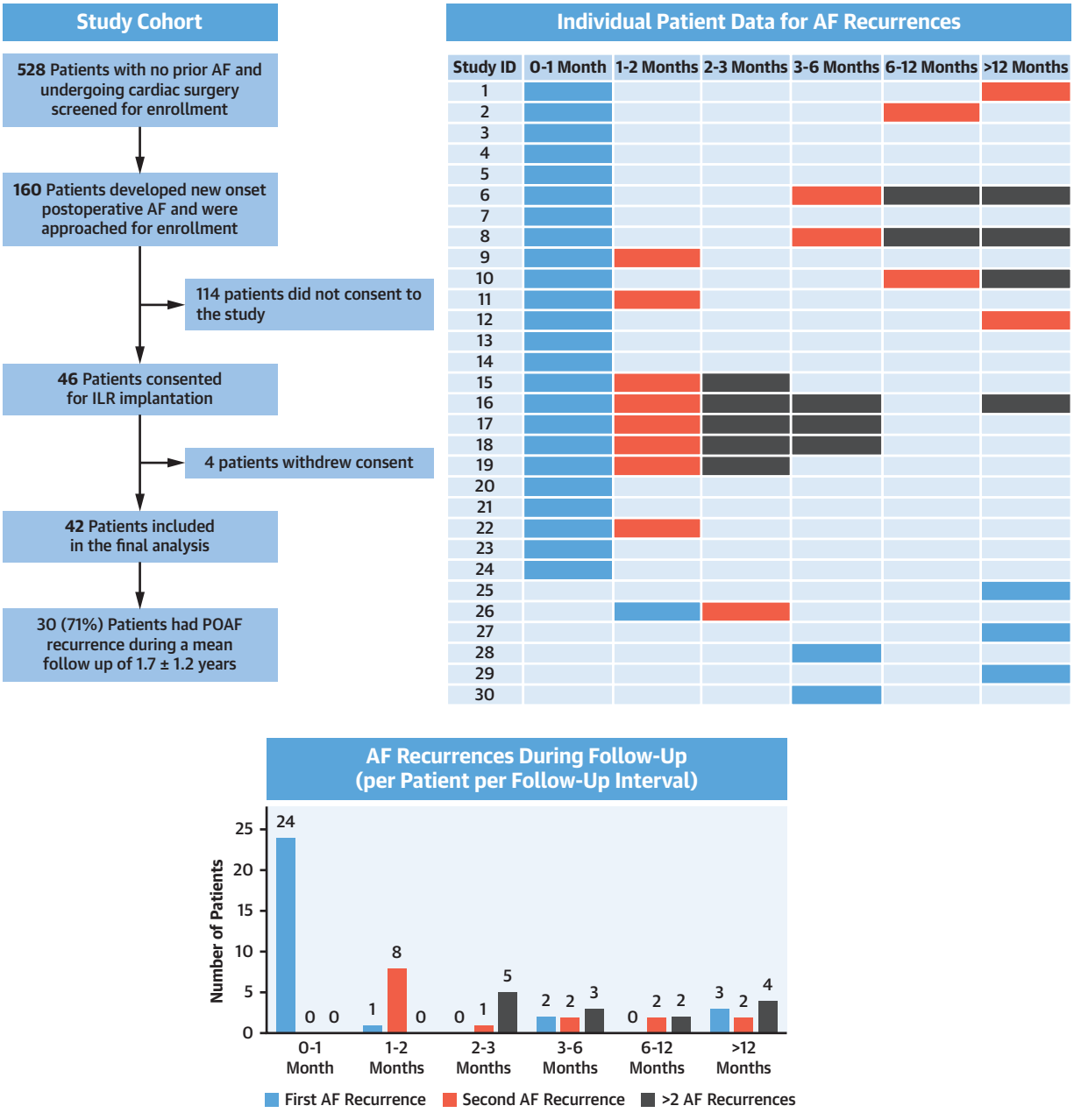
Values are mean ± SD, or n (%), or median (25% and 75% Interquartile range.

AF = atrial fibrillation; BMI = body mass index; BUN = blood urea nitrogen; CABG = coronary artery bypass graft; CAD: coronary artery disease; CHF = congestive heart failure; COPD = chronic obstructive pulmonary disease; HFrEF = Heart failure reduced ejection fraction (defined as LVEF ≤45%); INR = international normalized ratio; K = Potassium; LA = left atrium; LVEF = left ventricular ejection fraction; MI = myocardial infarction; Na = Sodium; OSA = obstructive sleep apnea; PAD = peripheral arterial disease; pBNP = pro brain natriuretic peptide; TSH = thyroid-stimulating hormone; WBC = white blood cells.

In our study, 33% of patients with AF at >12 months' follow-up were on OAC. Whether patients with POAF should remain on long-term anticoagulation is not clear in the current clinical and surgical guidelines due to lack of direct evidence. The rate of AF recurrence in our study is similar to the results reported by Park-Hansen et al (14), who found new AF recurrence in 16%, and additional repeated AF recurrence in 28% of patients after 3 months postsurgery. Such findings call into question the current guideline recommendations of anticoagulation for 30 days postoperatively.

In this study, 1 patient died (no recurrent AF noted) and 2 patients had a CVA (both had AF recurrence; 1 patient with AF duration <5 minutes and another with AF ≥5 minutes). Although our study was not powered to evaluate the association between new-onset POAF and these events, others have reported on such association. The EXCEL (Evaluation of XIENCE versus Coronary Artery Bypass Surgery for Effectiveness of Left Main Revascularization) trial (20) reported that new-onset POAF was an independent predictor of 3-year stroke with adjusted hazard ratio of 4.2 (1.74-10.11) in evaluating new-onset POAF post coronary revascularization. Further, a recent meta-analysis (21) encompassing 35 cohort studies (including cardiac and noncardiac surgeries), with more than 2.4 million patients reported that new-

CENTRAL ILLUSTRATION **Study Cohort and Postoperative Atrial Fibrillation Recurrences**



Abdelmoneim, S.S. et al. J Am Coll Cardiol EP. 2021;7(9):1134–1144.

(Left) Study cohort: Single-center, prospective observational study evaluating 42 patients undergoing cardiac surgery and diagnosed during index admission with new-onset, transient, POAF between May 2015 and December 2019. Before discharge, all patients received implantable loop recorders (ILR) for continuous monitoring. **(Right)** Individual patient data across follow-up time points for all AF recurrences (n = 30). **Blue color**: patients with first AF recurrence at any follow-up time point. **Red color:** patients with second AF recurrence after the first AF recurrence at any follow-up time point. **Gray color:** Patients with more than 2 AF recurrences at any follow-up time point. Bottom Bar graph extrapolated from the individual patient data, illustrating AF recurrence during follow-up (per patient per follow-up time points). Color coding and labels are shown in the graph. Twenty-four (80%) of the 30 patients had their first AF recurrence within the first month. For months 1 to 12 follow-up, 76% of patients had any AF recurrences: 3 (10%) had their first AF recurrence, 13 (43%) had their second AF recurrence, and 7 (23%) had more than 2 AF recurrences. Beyond 1 year of follow-up, 30% of patients had any AF recurrences: 3 (10%) had their first AF recurrence, 2 (7%) had their second AF recurrence, and 4 (13%) had more than 2 AF recurrences). AF= atrial fibrillation; POAF = postoperative atrial fibrillation.

onset POAF had a 62% higher odds of early stroke and a 44% higher odds of mortality compared with those without POAF (21).

Our study is unique in the sense of targeting the gaps in previous studies with its prospective design, inclusion of both CABG and valve surgeries, and the use of continuous rhythm monitoring for a prolonged period of close to 2 years. Hence, we were able to highlight the natural progression of POAF and its recurrence.

STUDY LIMITATIONS. Our study is limited by the single center, and small population studied, which limits the generalizability of our data. There are indeed existing challenges and limitations recruiting postoperative open heart surgery patients. Our study did not explore the precipitating factors for POAF development; however, it may be presumed that manipulation of the heart or surgical incision in the atrium may lead to abnormal atrial conduction precipitating POAF. Despite this, we did not find a significant difference in POAF between valve surgery and CABG, possibly due to the small numbers.

In general, the surgical team's protocol for stopping amiodarone postoperatively is 6 weeks. However, this was individualized at the discretion of the multidisciplinary team of cardiothoracic surgeon and cardiologist, which in turn could have influenced the outcomes of recurrence of AF. The decision of treatment with OAC may have been confounded with the nature of valve surgery and valve replaced, hence may have skewed the percentage of OAC and do not reflect the anticoagulation decision as it relates to AF recurrence.

CONCLUSIONS

Recurrence of AF in patients who develop transient POAF is common in patients undergoing open heart surgery with a high frequency of recurrence in the first month postoperatively (71% of patients, and of those, 59% had AF episodes equal or longer than 5 min; median AF duration at or beyond 1 month was 32 [IQR: 5.5-106] min and 15 [IQR: 6.3-49] min, respectively). However, for 1 to 12 months postoperative follow-up, 76% of patients had any AF recurrences, whereas beyond 1-year follow-up, 30% of patients had any AF recurrence, supporting the need for close outpatient follow-up. Further investigation into a strategy of a continuous ambulatory monitoring to determine the need for continued anticoagulation beyond the 30 days recommended by current clinical and surgical guidelines is warranted based on the findings of this study, within the context of AF duration beyond 1 month was short (<30 min).

Unanswered questions remain about the optimal anticoagulation duration. Further evidence is needed to determine the appropriate duration of anticoagulation in patients with postoperative AF through research studies with larger sample size, continuous monitoring, and a randomized design to better understand whether empiric long-term OAC treatment alters long-term prognosis in patients with new-onset, transient, POAF.

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ADDRESS FOR CORRESPONDENCE: Dr John F. Heitner, Division of Cardiology, Weill Cornell Medicine, New York-Presbyterian Brooklyn Hospital, 506 Sixth Street, Brooklyn, New York 11215, USA. E-mail: john.heitner@gmail.com.

PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: This prospective study highlights the natural history and progression of POAF. New-onset POAF recurrence may continue subacutely and in the long term. This highlights that these arrhythmias are not situational, as was traditionally assumed. Prolonged monitoring is of value and such monitoring is readily available and should be recommended for postoperative patients presenting with new-onset POAF.

TRANSLATIONAL OUTLOOK: The potential challenge to the current OAC recommendation for POAF after cardiac surgeries needs to be addressed by a guidelines committee. Future studies recommended to ultimately determine whether empiric long-term OAC treatment alters long-term prognosis of these patients. This should be investigated within the context of our finding that AF duration was <30 minutes beyond 1 month and the benefit of long-term OAC in the setting of brief AF episodes is still unclear.

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